

BAI-free ideal amenability when the quotient is amenable

A substantial partial answer to arXiv:math/0610137

Autonomous proof packet, lane 14

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Status. `candidate_partial_likely_valid`. The unrestricted source question is not claimed solved.

1 Source question

The source is M. Eshaghi Gordji, B. Hayati, and S. A. R. Hosseiniun, “Derivations into duals of closed ideals of Banach algebras,” arXiv:math/0610137v2 (2006) [1]. Their Theorem 5.4 says that if I is a closed ideal of a Banach algebra A , I has a bounded approximate identity (BAI), and both I and A/I are ideally amenable, then A is ideally amenable. On PDF page 13 they ask:

Is the above theorem valid if I does not have any bounded approximate identity?

$$\leq \inf\{\|j + i\| : i \in I \cap J\} \\ = \|j + J \cap I\|.$$

So, ψ is bounded. By open mapping theorem ψ is a homeomorphism and consequently $(\frac{J}{J \cap I})^* \cong (\frac{J+I}{I})^*$. Therefore there exists $\Phi_2 \in (I \cap J)^\perp$ such that $\tilde{D} = \delta_{\Phi_2}$. It shows that $D = \delta_{\Phi_1 + \Phi_2}$; D is inner and $\mathcal{A}$ is ideally amenable. ■

It is notable that the above theorem is true for weak amenability and amenability even if I does not have any bounded approximate identity [D].

Now we pose an open problem in this direction.

Question. Is valid the above theorem, if I does not have any bounded approximate identity ?

REFERENCES

- [B] A. Blanco, *On the weak amenability of $A(X)$ and its relation with the approximation property*, J. Functional analysis, 203 (2003) 1-26.
- [B-C-D] W. G. Bade, P. G. Curtis and H. G. Dales, *Amenability and weak amenability for Beurling and Lipschitz*

2 Definitions and notation

If X is a Banach A -bimodule, its dual actions are fixed by

$$\langle x, a \cdot \phi \rangle = \langle xa, \phi \rangle, \quad \langle x, \phi \cdot a \rangle = \langle ax, \phi \rangle.$$

A derivation $D : A \rightarrow X$ satisfies $D(ab) = a \cdot D(b) + D(a) \cdot b$; it is inner if $D(a) = a \cdot \phi - \phi \cdot a$ for some $\phi \in X$. The algebra A is *ideally amenable* if every derivation $A \rightarrow J^*$ is inner for every closed

two-sided ideal J of A . It is *amenable* in Johnson's sense if every derivation into the dual of every Banach A -bimodule is inner [2].

For closed ideals $I, J \triangleleft A$, write

$$K = I \cap J, \quad E_J = \overline{IJ + JI}^{\|\cdot\|} \subseteq K.$$

3 Proof intuition

Ideal amenability of I lets us remove, by an inner perturbation, the part of a derivation $D : A \rightarrow J^*$ visible on $K = I \cap J$. This initially gives only that $D(i)$ annihilates K for $i \in I$. Applying the derivation identity to products in I upgrades this to $D(I^2) = 0$, and weak amenability of I makes I^2 dense in I .

The adjusted derivation therefore descends to A/I . Its exact coefficient module is not usually the dual of a quotient ideal: it is $(J/E_J)^*$, where $E_J = \overline{IJ + JI}$. This explains the open problem. Ordinary amenability of A/I controls every dual module and immediately finishes the argument. Alternatively, if $E_J = I \cap J$ and $I + J$ is closed, then $J/E_J \cong (I + J)/I$ is a closed ideal of A/I , so ideal amenability suffices. The BAI in the source theorem supplies exactly these two structural properties. What remains open is whether endpoint ideal amenability supplies some different way around the residual module.

4 The BAI-free theorem

Theorem 1 (Amenable quotient). *Let I be a closed two-sided ideal of a Banach algebra A . If I is ideally amenable and A/I is amenable, then A is ideally amenable. No approximate identity is assumed on I .*

Proof. Let $J \triangleleft A$ be closed and let $D : A \rightarrow J^*$ be a continuous derivation. Put $K = I \cap J$, and let $r : J^* \rightarrow K^*$ be restriction. Since K is a closed ideal of I and r is an I -bimodule morphism, $rD|_I : I \rightarrow K^*$ is a derivation. Ideal amenability of I gives $\lambda \in K^*$ such that

$$rD(i) = i \cdot \lambda - \lambda \cdot i \quad (i \in I).$$

Choose a Hahn–Banach extension $\Lambda \in J^*$ of λ , and set $D_0 = D - \delta_\Lambda$. Then

$$D_0(i)|_K = 0 \quad (i \in I). \tag{1}$$

For $a, b \in I$ and $j \in J$, the dual actions and the derivation identity give

$$\langle j, D_0(ab) \rangle = \langle ja, D_0(b) \rangle + \langle bj, D_0(a) \rangle = 0,$$

because $ja, bj \in I \cap J = K$. Hence $D_0(I^2) = 0$. Ideal amenability of I implies weak amenability. Weak amenability implies $\overline{I^2} = I$: indeed, if $0 \neq f \in I^*$ annihilated I^2 , then $T(x) = f(x)f$ would be a nonzero derivation $I \rightarrow I^*$; it cannot be inner, because the bilinear form $\langle y, T(x) \rangle = f(x)f(y)$ is symmetric, whereas the form associated to an inner derivation is antisymmetric. Thus continuity yields

$$D_0|_I = 0. \tag{2}$$

Let $E = E_J = \overline{IJ + JI}$. For $i \in I$ and $a \in A$,

$$0 = D_0(ia) = i \cdot D_0(a), \quad 0 = D_0(ai) = D_0(a) \cdot i.$$

Pairing with elements of J shows that $D_0(a)$ annihilates JI and IJ , respectively. Therefore

$$D_0(A) \subseteq E^\perp \cong (J/E)^*.$$

The closed space E is an A -submodule of J , and I acts trivially on J/E . Consequently J/E is a Banach (A/I) -bimodule. By (2),

$$\bar{D} : A/I \longrightarrow (J/E)^*, \quad \bar{D}(a + I) = D_0(a),$$

is a well-defined continuous derivation. Amenability of A/I makes $\bar{D}$ inner, implemented by some $\mu \in (J/E)^* = E^\perp \subseteq J^*$. Thus $D_0 = \delta_\mu$, and $D = \delta_{\Lambda + \mu}$ is inner. Since J and D were arbitrary, A is ideally amenable. $\square$

5 The exact structural criterion

Theorem 2 (Closed-sum/essential-intersection criterion). *Let $I \triangleleft A$ be closed. Suppose I and A/I are ideally amenable and, for every closed ideal $J \triangleleft A$,*

$$I + J \text{ is closed,} \quad I \cap J = \overline{IJ + JI}. \quad (3)$$

Then A is ideally amenable.

Proof. Repeat the proof of Theorem 1 through the factorization

$$\bar{D} : A/I \longrightarrow (J/E_J)^*.$$

By (3), $E_J = I \cap J$. The natural map

$$\Theta : J/(I \cap J) \longrightarrow (I + J)/I, \quad j + (I \cap J) \longmapsto j + I,$$

is a continuous bijective bimodule map. Its domain is Banach, and its range is a closed ideal of A/I because $I + J$ is closed. The open mapping theorem makes Θ a topological bimodule isomorphism. Hence $(J/E_J)^*$ is isomorphic, as an (A/I) -bimodule, to the dual of the closed ideal $(I + J)/I$. Ideal amenability of A/I makes $\bar{D}$ inner. The last step of Theorem 1 completes the proof. $\square$

Proposition 3 (Recovery of the source theorem). *If I has a bounded two-sided approximate identity, then (3) holds for every closed ideal $J \triangleleft A$.*

Proof. Let (e_α) be a BAI for I with $\sup_\alpha \|e_\alpha\| \leq M$. For $k \in I \cap J$, $e_\alpha k \rightarrow k$ and $e_\alpha k \in IJ$; hence $I \cap J = \overline{IJ} \subseteq \overline{IJ + JI}$, while the reverse inclusion is automatic.

To prove $I + J$ closed, first note that for $j \in J$, $i \in I$, and $\varepsilon > 0$, one may choose α so that $\|i - e_\alpha i\| < \varepsilon$. Since $e_\alpha j \in I \cap J$,

$$\text{dist}(j, I \cap J) \leq \|j - e_\alpha j\| \leq (1 + M)\|j - i\| + \varepsilon.$$

Taking the infimum over i and then $\varepsilon \downarrow 0$ shows that the injective map $J/(I \cap J) \rightarrow A/I$ is bounded below. Its range $(I + J)/I$ is therefore closed. Thus $I + J$ is closed. $\square$

Corollary 4. *Suppose I is topologically simple and ideally amenable, A/I is ideally amenable, and the image in A/I of every closed ideal of A is closed (in particular, this holds when A/I is finite-dimensional). Then A is ideally amenable, without a BAI assumption on I .*

Proof. For each closed $J \triangleleft A$, topological simplicity gives $I \cap J = 0$ or I . In the first case $E_J = 0$. In the second, $\bar{I}^2 = I$ by ideal amenability, so $E_J = I$. Thus $E_J = I \cap J$. Closedness of the image $(I + J)/I$ implies closedness of $I + J$, and Theorem 2 applies. $\square$

6 A genuine non-BAI example

Let H be an infinite-dimensional Hilbert space and let $\mathcal{S}_1(H)$ be the trace-class operators with composition and trace norm. This algebra is topologically simple: if a nonzero closed ideal contains $T \neq 0$, rank-one operators on the left and right sandwich T to produce every rank-one operator, and finite-rank operators are trace-norm dense.

It is also weakly amenable. Given a derivation $D : \mathcal{S}_1(H) \rightarrow \mathcal{S}_1(H)^* = \mathcal{B}(H)$, fix a unit vector e and write $\theta_{x,e}z = \langle z, e \rangle x$. The formula

$$Sx = -D(\theta_{x,e})e$$

defines a bounded $S \in \mathcal{B}(H)$. From $a\theta_{x,e} = \theta_{ax,e}$ and the derivation identity, evaluated at e , one obtains

$$D(a)x = aSx - Sax \quad (a \in \mathcal{S}_1(H), x \in H).$$

Thus $D(a) = aS - Sa$ is inner. Topological simplicity now makes $\mathcal{S}_1(H)$ ideally amenable, since its only closed ideals are 0 and itself.

On the other hand $\mathcal{S}_1(H)$ has no bounded approximate identity. If (u_α) were one with $\|u_\alpha\|_1 \leq C$, choose orthonormal $e_1, \dots, e_N$ and the rank-one projections p_k onto $\mathbb{C}e_k$. For one α sufficiently far out, $\|u_\alpha p_k - p_k\|_1 < 1/2$ for all $k \leq N$, so

$$C \geq \|u_\alpha\|_1 \geq \sum_{k=1}^N |\langle u_\alpha e_k, e_k \rangle| > N/2,$$

a contradiction for large N . Therefore Theorem 1 applies non-vacuously to the unitization $\mathcal{S}_1(H)^\#$, whose quotient by $\mathcal{S}_1(H)$ is $\mathbb{C}$.

7 Why this does not settle the full question

The common reduction produces $(J/E_J)^*$, not automatically the dual of a closed ideal of A/I . Endpoint ideal amenability alone is therefore not enough for the proof given here. Eight focused upgrades tested whether $E_J = I \cap J$ or closedness of $I + J$ follows cohomologically, audited a later claimed full theorem, and explored module-extension counterexamples. No proof of either missing relation and no counterexample was found.

Ranjbari and Rejali's Theorem 4.1 [3] states a stronger Fréchet-algebra theorem that would specialize to the full Banach result. Its proof first treats an element of $(I \cap J)^*$ as evaluable on arbitrary elements of J . Repairing that step gives the valid conclusion $D_0(I^2) = 0$ used above, but the final appeal to the weak-amenability three-space proof still does not identify $(J/E_J)^*$ as an ideal dual. The exact audit is recorded in the companion proof-gap packet. Accordingly, the unrestricted question remains open in this investigation.

8 Verification and novelty

The source statement, all module actions, the density-of-products step, both annihilator calculations, and the quotient-module identification were audited explicitly. The BAI recovery and trace-class example are proved in the packet rather than inferred from analogy. No computation is used.

The run indexes, exact-title searches, close keyword searches, and the later 2017 primary PDF were checked. No inspected source stated the two partial theorems above with a valid proof. Novelty confidence is moderate; exhaustive priority is not claimed. Human review should focus on (1)–(2) and on the topological identification in Theorem 2.

References

- [1] M. Eshaghi Gordji, B. Hayati, and S. A. R. Hosseiniun, *Derivations into duals of closed ideals of Banach algebras*, arXiv:math/0610137v2 (2006).
- [2] B. E. Johnson, *Cohomology in Banach algebras*, Memoirs of the American Mathematical Society 127 (1972), 1–96.
- [3] A. Ranjbari and A. Rejali, *Ideal Amenability of Fréchet Algebras*, U.P.B. Scientific Bulletin, Series A 79(4) (2017), 51–60.